Prelims Introductory Calculus MT 2026: Sheet 7

1. Find the gradient $\boldsymbol{\nabla} f$ for each of the following functions:

   (a) $f(x,y) = e^{x^2-y^2}\sin 2xy$; (b) $f(x,y,z) = (x^2+y^2+z^2)^{-1/2}$.

2. For each of the following functions sketch some contours $f(x,y) =$ constant (include enough contours to explore all four quadrants of (x,y)) and indicate $\boldsymbol{\nabla} f$ by arrows at some points on these curves. Use one set of axes for each case.

   (a) $f(x,y) = xy$; (b) $f(x,y) = \frac{1}{4}x^2 + y^2$; (c) $f(x,y) = \frac{x}{y}$.

3. Show that there is no function $f(x,y,z)$ such that $\boldsymbol{\nabla} f(x,y,z) = (y,z,x)$.

4. Let $f(x,y) = x^2y^3$ and determine $\boldsymbol{\nabla} f$. Given the vector $\mathbf{a} = (1,1)$ and the unit vector $\mathbf{v} = (a,b)$ with $a^2+b^2 = 1$, evaluate

   $$\lim_{t\to 0} \frac{f(\mathbf{a}+t\mathbf{v}) - f(\mathbf{a})}{t}.$$

   What is the maximum value of this limit that can be obtained by varying a in the vector $\mathbf{v}$?

5. A bug is walking on the xy-plane which is has a toxic environment. The level of toxicity is given by the function $f(x,y) = 2x^2y - 3x^3$. The bug starts at the point $(1,2)$. In what direction away from $(1,2)$ should it initially move in order to lower its exposure to the toxin as rapidly as possible?

6. Find a unit vector perpendicular to the surface $x^2y + y^2z + z^2x + 1 = 0$ at the point $(1,2,-1)$. Hence, or otherwise, write the equations of the tangent plane and the normal line at this point.

7. Let f and g be sufficiently differentiable functions of x, y, z.

   (a) Prove the following:

   i. $\boldsymbol{\nabla}(fg) = f\boldsymbol{\nabla} g + g\boldsymbol{\nabla} f$;

   ii. $\boldsymbol{\nabla}(f^n) = nf^{n-1}\boldsymbol{\nabla} f$;

   iii. $\boldsymbol{\nabla}\left(\frac{f}{g}\right) = \frac{g\boldsymbol{\nabla} f - f\boldsymbol{\nabla} g}{g^2}$.

   (b) In Cartesian coordinates (x,y,z) the Laplacian operator, $\boldsymbol{\nabla}^2$, is defined by

   $$\boldsymbol{\nabla}^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}.$$

   Show that

   $$\boldsymbol{\nabla}^2(fg) = f\boldsymbol{\nabla}^2 g + 2(\boldsymbol{\nabla} f)\cdot(\boldsymbol{\nabla} g) + g\boldsymbol{\nabla}^2 f.$$

8. Use the Taylor series for a function of two variables to expand the following functions about the point $x = 1, y = 1$, to the order shown.

   (a) $f(x,y) = x^3 + y^2 + 3x^2y$, to all orders;

   (b) $f(x,y) = x^2 + y + \cos(2\pi xy)$, to second order.